Gender at the Manhattan Project: A Historicised Account of Identity and Epistemic Injustice in Physics

Modern Sciences

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'Los Alamos is about 7200 feet above sea level... I happened to feel fine, but the altitude affected my cooking and baking. It took much longer for water to boil, so my first few meals were on the tardy side. Cakes rose beautifully in the oven, but fell magically the moment they were removed. I'd never enjoyed baking anyhow, so I now had an excuse not to bake at all.'[1]

[1] Marshak, 1988, 8-9.

Introduction

I aim to develop a historicised account of women's engagement within physics by examining gender and identity at the Manhattan Project. Scholarly understanding of women's underrepresentation in physics is limited.[2] Despite many shifts within physics culture, from the influx of weapons-based research to the rise of computer technology, and increased numbers of women in academia generally, proportions of women in physics have stagnated within the last few decades.[3] I wish to illuminate this phenomenon by demonstrating similarities (and differences) regarding gender and physics at the Project versus in the United States today. I have looked empirically at autobiographical material produced by women present across various sites of the Project, using modern analytic resources to explore their self-created narratives of gendered experiences and their relationships to science. These resources include physics education research (PER), gender and feminist theory, and conceptions of epistemic injustice, which harms an individual in their capacity as a knower. I will argue via multiple examples while gender roles and physics cultures differed at the Project, as did the available linguistic and social tools, the same factors often enabled women to engage or disengage with physics. Showing through this specific historical episode women have been making significant contributions within the field for decades, with relatable modes of experience, I argue this account of identity and epistemic injustice at the Project will therefore provide insight into factors critical for creating environments and cultures which help women engage within the physical sciences.

The Manhattan Project

The Manhattan Project entailed the (successful) US-based effort to build an atomic bomb during World War Two. Its main sites were Clinton Engineer Works at Oak Ridge, Tennessee, comprised mainly of industrial facilities for separating uranium isotopes, Hanford, near Richland, Washington, where nuclear reactors for producing plutonium were based, and Los Alamos, New Mexico, where physicists and engineers came together for design and construction of the bomb. Also subsumed into the Project were various groups and specialists at universities across the US, and multiple companies.[4]

[2] Ito et al, 2016, 1.
[3] Ibid. For an account of changing physics culture, see Galison, 1997.
[4] Herzenberg and Howes, 1999, 9-10, 40, 73, 81-82, 140.

Why focus on the Manhattan Project? As a wartime endeavour, more women were recruited into STEM positions, given an increased number of roles and the relative absence of men. Security and budget considerations also made it optimal to hire wives of men already working on the Project. At Los Alamos, this was further incentivised as both spouses had to be employed to receive the best housing and household help. Most sites had female staff, including female scientists and technicians. For example, in 1944, 9% of employees at the Hanford site were women (approximately 4,000).[5] Women in STEM have historically been unlikely to create records of their achievements or experiences. However, patriotic duty, and an accompanying sense of pride, allowed these women external purpose to write about themselves while shifting the focus to achievement within a broader scope.[6] Thus, there is a relatively large body of autobiographical material related to the Project written by women.[7]

These autobiographies are often presented as attempts to give insight into other famous scientists. This presumably appealed to readers and publishers, and obscured any sense of impropriety in relating one's own experiences. For example, take Elizabeth Rona's 1978 book, *How It Came About*. Rona worked on the Project in a somewhat oblique role, partly because of her Hungarian background, using her significant experience in radiochemistry to undertake (for example) research into producing large quantities of pure polonium-210 and lead-210.[8] Rona tellingly related '[f]or a long time my friends have urged me to write about my scientific activities and about the distinguished scientists whom I have known and with whom I have been associated'; it was not *her* idea, and she was not writing about *herself*.[9] She had to be persuaded. The foreword reads '[o]f special interest is her personal acquaintance with most of the scientists involved in research on radioactivity.'[10] Again, we see emphasis not on Rona's scientific achievements, but on other scientists. This contributes to bias in this essay towards Los Alamos, with the collection of famous scientists at this site affecting the relative abundance of source material.

[5] Ibid., 2, 4, 13, 15, 18-20, 57-60, 82, 84, 101, 105, 132, 146. Weaver, 1999, viii.
[6] Wolkowitz, 2000, 246.
[7] Herzenberg and Howes, 17, 133, 170.
[8] Rona, 1978, 54-57.
[9] Ibid., 1.
[10] Pollard, 1978, ii.

Literature and Source Material Overview

One reason narratives about the Project are so focused on male physicists is precedent set by its official history, *Atomic Energy for Military Purposes*. Written by physicist Henry Smyth and commissioned by General Leslie Groves (director of the Project), it is highly technical and focuses on facts rather than experiences, with the purpose of revealing building an atomic bomb was possible.[11] Indeed, Smyth asserted this book was for 'engineers and scientific men who can understand such things' in America.[12] This helped create the image of the Project as a male venture into physics and engineering research that still resides in the public consciousness today.[13]

It is necessary to acknowledge Herzenberg and Howes' seminal work tracking down women who worked on the Project, utilising a wealth of personal interviews and questionnaires to write *Their Day in the Sun*. This book is incredibly important in demonstrating the great breadth of roles women occupied. Nevertheless, there is not much holistic analysis of autobiographical material created by women who worked on the Project. Some exists on gender and the Project as a weapon-building venture, such as Wolkowitz's chapter *Nuclear Families*. However, I intend to explore these sources with respect to the Project as a scientific endeavour.

This material varies significantly in length and publication dates, spanning various occupations, sites, and narratives. The different accounts, broadly categorised (with overlap) as by housewives, working wives, women in technical and scientific roles, and female physicists, tend to be concerned with different subject matter. Perhaps this is somewhat obvious, as people with different occupations do different things. For example, Charlotte Serber, Scientific Librarian at Los Alamos and wife of physicist Robert Serber, wrote '[f]or the potential working wife, there was one chief worry. Could she manage her home… and work too? Would her children become delinquents?'.[14] These are different 'chief' concerns to an unmarried physicist or housewife, reflected in their writing. Analysing the material broadly will, I hope, create a more comprehensive image of the complex relationships women had with

[11] Fermi, 1961, 180. Groves, 1945, v.
[12] Smyth, 1945, vi.
[13] Schwartz, 2008.
[14] Serber, 1988, 67. Herzenberg and Howes, 1999, 15.

science both as scientists and otherwise (e.g. as wives of scientists, or housewives who chose not to pursue scientific careers after doing related degrees).

I also acknowledge the role of the Los Alamos Historical Society in getting several accounts published. For example, one key source, *Standing By and Making Do*, contains short essays from various women who lived at Los Alamos. This was written in 1946 as they 'began to think of what they had accomplished and its possible historical significance'.[15] However, they were rejected by several publishers. It was 1987 when the Society selected the manuscript for publication. This perhaps indicates increasing social interest in the experiences of women. As a counterpoint, I have read *Behind Tall Fences*. Of the 21 accounts spanning 1941 to 1951, only one was written by a woman, yet it is not presented as a book about men's experiences at Los Alamos.[16] This accords with feminist viewpoints recognising men are considered neutral defaults of personhood while women are 'Other'.[17] Much of the material therein is highly technical; the accounts written by women provide fuller narratives on life at the site, although material written particularly by female physicists contains similar content.

Common Themes

Science is often presumed to be, and advocated as, 'value-free', yet physics has been historically masculinised both inherently and symbolically. This can make it difficult for women to participate while retaining their sense of feminine identity.[18] The American Institute of Physics has found women have comprised around 20% of individuals within physics, from undergraduate to faculty level, for about the last decade.[19] In the 1930s, around 3% of physics doctorates in the US were awarded to women.[20] Gender roles were more rigid and pervasive; women's exclusions from the field were often more explicit. Perhaps one might therefore assume, given significant cultural changes within physics and US society, much has changed for women in physics. Within this section, however, I aim to demonstrate how several specific themes identified in modern research for women's increased engagement in physics are visible within the source material, while highlighting historical contingencies.

[15] Lyon, 1988, vi.
[16] Allred, 1996, xiii.
[17] E.g. de Beauvoir, 1953, 14-21, 29.
[18] Blue et al, 2016, 4, 7. Ito et al, 2016, 8. Ong, 2005, 597-598.
[19] Staley, 2023, 3.
[20] Herzenberg and Howes, 1999, 35.

Thematic Analysis

Modern PER has identified various contextual factors increasing likelihood women remain within physics. I highlight childhood influences and father figures, inspiring teachers and role models, confidence in ability, a love of nature, participation in activities which increase a sense of belonging, and devotion to one's work. I broadly categorise the first two as external factors, and the latter four as internal. The internal factors are influenced by context and environment, just less directly.

Ideally, this would result in a positive account of strategies women who persevere in the sciences employ, and environments allowing them to do so.[21] However, I do not wish to suggest women must attempt to emulate these factors to succeed within the physical sciences. This consists within what Lugones has identified as 'multiplicity', wherein an individual engages in 'performances of superiority' as part of manipulation and utilisation of stereotypes to allow them to retain traits and behaviours exhibited with their various identities.[22] This should not encourage another practice identified by Lugones, 'fragmentation', which entails methods and modes by which an individual separates components of their identity to better fit within their community and 'approximate ordinariness'. For example, consider a woman who avoids wearing pink within the laboratory, having (subconsciously or otherwise) both associated this with feminine identity and at odds with traits or behaviour 'acceptable' within her scientific community.[23] Fragmentation and multiplicity can force students to compromise feminine identities for scientific identities, and solidify gendered boundaries within science.

Common factors do not signal only these women are or were able to participate within physics, and I have only been able to locate source material from a small percentage of women involved in the Project.[24] Borrowing from Herzenberg and Howes, '[t]he women whose experiences are described here must stand in for a number of others about whom we have not been able to learn as much'.[25] However, understanding what has often helped women engage with physics, historically and currently, may allow us to change practices and environments so women do

[21] Ong, 2005, 594.
[22] Ibid., 600-601, 603.
[23] Ibid., 593-595.
[24] Wolkowitz, 2000, 235.
[25] Herzenberg and Howes, 1999, 100, 155.

not feel – subconsciously or otherwise – they have to. It is the field, its culture, and our perceptions that should change if more women are to succeed within the physical sciences.[26]

External Factors

In this subsection, I largely focus on experiences prior to the Project linked to women's eventual involvement. One factor found to indicate higher likelihood for women engaging with physics is having had a father figure in a relevant field of research who encouraged them to take science classes.[27] This is born out in Rona's self-created personal narrative, as she wrote

> '[m]y interest in science, I think, had its inception in my early childhood… one day… I looked out of my bedroom window… and saw my father writing, deeply involved in something, oblivious of the outside world. "What can be so interesting that it makes it worthwhile to get up at the crack of dawn?" I asked my father. He cryptically answered, "research". I think that my interest in scientific research started unconsciously at that early age'.[28]

Rona's father was one of the first physicians to investigate radium's medical applications in Hungary.[29] Rona 'had always wanted to study medicine… but my father opposed my choosing it as a career. He thought it was too hard for a woman. After his untimely death, I respected his intentions and chose to pursue a career in chemistry and physics'.[30] Rona thus denotes this factor as key to her engagement within the physical sciences. This also demonstrates changing gender relations to science. Whereas US undergraduate enrolment of women in medicine is now higher than for men, Rona grew up with medicine being considered 'too hard for a woman'. Chemistry and physics, it seems, were relatively acceptable – Rona makes no reference to resistance to this educative and career path.[31]

It is arguably problematic to emphasise the influence of men on women in the sciences: perhaps having a scientific mother figure is similarly important. The most famous dynasty in radioactivity suggests this – Marie Curie, her daughter Irène, and her granddaughter Helen all

[26] Blue et al, 2016, 4. Sax, 2016, 5. Staley, 2023, 13.
[27] Blue et al., 2016, 3.
[28] Rona, 1978, 1-2.
[29] Herzenberg and Howes, 1999, 88.
[30] Rona, 1978, 2.
[31] AAMC, 2022.

worked as nuclear scientists.[32] The picture is complicated by how many female scientists at the Project were either single or married to scientists (which I discuss later). This remains an open question.

Another factor entails positive educational experiences, and inspiring teachers.[33] Rona was complimentary of all her main research supervisors, discussing the 'pleasant' atmospheres at various institutions where she was a student.[34] High school physics courses which emphasise understanding over memorisation are significant in engaging women at undergraduate level in physics.[35] Joan Hinton, who worked as a physics graduate student at Los Alamos, relayed her fascination with atomic physics began when her high school chemistry teacher showed her a cloud chamber.[36] Edith Hinkley Quimby similarly related her high school teacher for physics and chemistry was a key influence in her choice to major in physics and mathematics (being the first woman to do so at Whitman College).[37] Quimby went on to work within the Project at Columbia and Oak Ridge, in medical physics. Rona later became a chemistry teacher at the woman-only Trinity College in Washington D.C., and wrote:

> '[a]nother difficulty was that the students had been assigned a text-book to use during the course. They were unhappy that my lectures did not follow this text. Some of the students who had received A+ grades before were lost and frustrated. Fortunately, they got used to this new style, and discussion and laboratory periods helped. One very reluctant and rebellious student became interested, and after graduation she told me that I was an inspiration to her'.[38]

This clearly accords with modern research findings on the importance of understanding over memorisation.

Women did not always have such positive experiences. Eleanor Ewing Ehrlich, who worked on implosion calculations at Los Alamos, was enrolled in an undergraduate physics class at the University of Illinois. Upon first walking into the classroom, the only woman in the class, her teacher told her "Miss Ewing, we do not mix sexes in physics. We have the girls' row down in

[32] Rona, 1978, 26, 34.
[33] Ito et al, 2016, 1-2.
[34] Rona, 1978, 3, 8-9, 12-13, 15-16.
[35] Blue et al., 2016, 3.
[36] Herzenberg and Howes, 1999, 51.
[37] Ibid., 115-117.
[38] Rona, 1978, 56-57.

front, please".[39] While Ehrlich completed an M.A. in mathematics, it is understandable why this sort of gendered isolation (nowadays less explicit) would have been highly discouraging. Leona Marshall Woods Libby, who worked as a nuclear physicist at all three of the main Project sites, similarly expounded upon influential interactions within her book *The Uranium People*. She wrote how prior to the War,

> 'when I got up the courage to ask [physicist James] Franck if he would allow me to be his graduate student… he accepted me. But, he said, he must give me a warning… when he was a graduate student [h]is professor had said to him, "You are a Jew and you will starve to death." So he had to say to me "You are a woman and you will starve to death." I looked at his corpulence; with not the least indication of any starvation in his life – and decided that I could not tolerate such a depressing idea… So that was the end of that, although his students loved him deeply and held him in great esteem. I worked instead for Robert Mulliken, who had a pleasant attitude, neither encouraging nor discouraging me'.[40]

Clearly, Libby disengaged with Franck as a supervisor based on his gendered attitude towards her, whether one interprets this as sexism or otherwise. Her comment on Mulliken is also revealing as to what atmosphere Libby felt conducive to her own research – a sort of academic neutrality. Overly pandering to women in the sciences, and ultimately treating them differently, can be condescending.

Internal Factors

A 2016 historical survey of female physics majors in the US over the last four decades, compared to women in other STEM fields, indicates a number of recurring character traits: confidence in mathematical ability, valuing college as a learning opportunity, planning to attend graduate school, and desiring to make theoretical scientific contributions.[41] Herzenberg and Howes identify a slightly different list of traits common to scientists at the Project: '[m]ale or female, they possessed considerable intelligence, scientific training, and physical energy, and in many cases a notable dislike of regimentation.'[42]

[39] Herzenberg and Howes, 1999, 108.
[40] Libby, 1979, 29-30.
[41] Sax, 2016, 5. In Staley, 2023, 13.
[42] Herzenberg and Howes, 1999, 51.

I associate confidence with what Montmarquet categorises as the epistemic virtue of 'intellectual courage', or 'the willingness to conceive and examine alternatives to popularly held beliefs, perseverance in the face of opposition from others (until one is convinced one is mistaken), and the determination required to see such a project through to completion'.[43] Somewhat extending mathematical to scientific ability, in the realm of scientific research, intellectual courage plausibly entails confidence in one's scientific abilities. There is certainly evidence of self-assuredness within the sources. Herzenberg and Howes discussed the female physicist Chien-Shiung Wu, who helped identify Xenon-137 in the Hanford reactors as a so-called 'reactor poison' which brings fission processes to a halt: '[f]ellow graduate students describe her as extremely competent and very independent. If offered help with an experiment, she would announce that she preferred to do it herself'.[44] Libby, discussing her time as a graduate student working on the first atomic pile at the University of Chicago, wrote '[o]ne day, [Enrico] Fermi [Libby's supervisor] asked me what I thought of his book on thermodynamics… [b]eing young and brash, I told him that it did very well as a baby book'.[45] Not every graduate student would convey this opinion to their world-leading supervisor. Libby also described how she would often 'instinctively chose [the] more difficult problems' set by Fermi while they worked together at Los Alamos, and 'failed to solve only a remarkably small number of problems'.[46] This suggests improving women's intellectual courage and epistemic confidence should be a key focus within PER.

Rona's description of her early years emphasises her 'love of nature', although she does not link this explicitly to her career path.[47] Rona also appreciated this trait in fellow nuclear scientist, Irène Curie.[48] Libby's attitude towards flowers differs somewhat to Rona's: 'hydrangea is more or less as beautiful as cabbage'.[49] However, Libby made frequent reference to her enjoyment of nature, for example when hiking in Los Alamos.[50] Jean Bacher, who did calculations for the Theoretical Physics Division, related physical activity with love of nature and eagerness to learn by relating how at the site she gained 'that persistent curiosity about

[43] Montmarquet, 1993, 23. In Fricker, 2007, 49-50.
[44] Herzenberg and Howes, 1999, 45-46.
[45] Libby, 1979, 151.
[46] Ibid., 148.
[47] Rona, 1978, 2.
[48] Ibid., 27, 33.
[49] Libby, 1979, 29.
[50] Ibid., 15, 204-206.

what lies ahead and at the top' when hiking.[51] Herzenberg and Howes referred to this quality more generally as 'their vigor, their enthusiasm, and their interest in the world'.[52]

I will now address the role of belonging, defined by Ito et al. (2016) as 'the extent to which individuals feel like… [and] the extent to which students subjectively perceive that they are valued, accepted, and legitimate members in their academic domain'.[53] Certain behaviours and traits, I argue, allowed women at the Project to increase belonging, which suggests they were better able to thrive as scientists. The Project as a patriotic wartime venture apparently inspired a general sense of camaraderie.[54] Particularly focusing on Los Alamos, several sources refer to the importance of hobbies like 'skiing and horseback riding, mountain climbing and folk dancing'.[55] Herzenberg and Howes, using personal interviews, outline how 'Hinton's intelligence and physical daring well qualified her for this group. She also played violin in a quartet that included [physicists] Edward Teller and Otto Frisch'.[56] Here we have a reference to Hinton's qualities and activities that allowed her to 'belong'.

Many of these women demonstrated similar enthusiasm for scientific research. Rona, choosing a postgraduate supervisor, was advised one (George Bredig) would allow her to publish a paper, and the other (Kasimir Fajans) her to enter a 'new and exciting field (radioactivity)'. Rona chose the latter.[57] Several female physicists who worked on the Project, including Libby, are reported to have been back at work as soon as possible after having given birth, although the influence of lack of childcare is debatable.[58] Libby hid her pregnancy from colleagues as her supervisor 'would probably have insisted on kicking me out of the reactor building', and was working until two days before the birth. Rona paid particular attention to scientific devotion within a chapter entitled 'Great Scientists', and wrote of Marie Curie '[Curie's] life consisted of long, assiduous, unbroken, labor without any hope of immediate distinction or reward; simply the pure love of the work itself, much toil solely in the interest of pure science'.[59] Rona described Lise Meitner, the celebrated nuclear physicist whom she worked under in Berlin-

[51] Bacher, 1988, 111. Herzenberg and Howes, 1999, 98-99. Serber and Wilson, 1988, 102.
[52] Herzenberg and Howes, 1999, 3.
[53] Ito et al, 2016, 1-2.
[54] E.g. Mark, 1988, 40.
[55] Marshak, 1988, 9. See also Bacher, 1988, 103, Barnett, 1988, 95, 106.
[56] Herzenberg and Howes, 1999, 55.
[57] Rona, 1978, 3.
[58] Herzenberg and Howes, 1999, 42, 50, 58. Libby, 1979, 164. Wolkowitz, 2000, 244.
[59] Rona, 1978, 49-50.

Dahlem, in similar terms: 'one had to admire her devotion to her work and her critical search for the right solution to scientific problems.'[60] Rona's work at the Project illustrates her own devotion to scientific ideals in the form of uncompensated labour and sacrifice of intellectual property. After solving a particular 'problem', Rona 'was asked if I would give away my method without selling it. I gave all the details without compensation'.[61]

Passion Versus Pragmatism

As opposed to devotion, multiple accounts reflect a certain pragmatism around scientific and technical roles. Many women discuss their occupations in ways suggesting they do not find being a 'woman in STEM' particularly shocking or unlikely in terms of ability. For example, Ellen Weaver was a scientist working in the chemistry labs at Oak Ridge, Tennessee, undertaking work such as quantitative analysis of fission products from the graphite pile. Her self-narration in the foreword of *Their Day in the Sun* reads 'I had majored in chemistry in college, not out of any love of science but because I wanted a skill that would empower me to support my husband, a physicist, through graduate school'.[62] A woman now may be less likely to admit this, as we celebrate 'women in STEM' for their devotion and independence. Perhaps this is one way emphasising passion and devotion to science as necessary qualities for success puts too much pressure on women, and discourages them from pursuing scientific education and careers.

Unusually, women at the Project were sometimes actively pressured into working, with many not having originally intended to, and hiring processes for wives were relatively simple.[63] Some women suggested they were not working because they prioritised other things, like raising their children.[64] Of course, these prioritisations are related to gender roles, but I would like to avoid condescension towards historical agents in implying being a housewife rather than a physicist was always undesirable. Eleanor Jette, who studied mathematics at university, wrote a short book on her time at Los Alamos, where she moved with her physicist husband Eric. She related

[60] Ibid., 10-11, 13.
[61] Ibid., 57. See Wellerstein, 2008 for a discussion of property rights and patenting as an instrument of control in the Manhattan Project.
[62] Weaver, 1999, vii.
[63] Herzenberg and Howes, 1999, 172. Serber, 1988, 57.
[64] Wolkowitz, 2000, 244.

> ‘I, too, was technically educated… I loved to play mathematical games… Once, during my academic career, I had a couple of unused hours. To fill them, I took a course on the theory of probability. It had a high-sounding name but a very practical application. It enabled me to figure the odds on certain combinations of cards coming up in card games like bridge and poker… [and] in horse races.’[65]

Again, we see a pragmatic component to Jette’s motivations in studying a STEM subject, without any suggestion of lack of ability. Jette recounted she was unsure, before arriving at the site, whether she would want to work in the so-called Technical Area or not.[66] However, her concern for her ‘disoriented’ son and her ‘pent-up weariness, worry, loneliness, and resentment’ meant she decided (emphatically) not to work. [67] Her attitude that ‘the wives, who should be at home taking care of the children, were intimidated or encouraged to work’ contrasts quite starkly to common opinion today on where women – classed as ‘wives’ – ‘should be’.[68] A quotation from Bernice Brode, who did calculations for the Theoretical Physics Division at Los Alamos, well summarises the attitudes of several working women, relating when the Women’s Auxiliary Core arrived, ‘wives were permitted to quit and only a few of the wives without children stayed on’.[69] It seems part of this phenomenon was a lack of suitable childcare, which continues to preclude women from working in STEM today.[70]

Jette provided an account of her husband’s attempt to soothe her as ‘[h]onest, darling, I didn’t have any idea you were cleared to work in the Tech Area. I don’t want you to work in the Tech Area. I’ve got enough troubles without coming home to a cold dinner.... No woman who can cook like you do has any business working in the Tech Area’. Jette’s husband’s attempted placation (or how Jette narrated it) has certainly dated. It is difficult, and nuanced, to analyse this passage and acknowledge Jette may not have had the social tools to make sense of harms perpetuated by women’s exclusions from STEM (say, on the basis of childcare), while respecting her agency to prefer being a housewife and feel comforted by appeals to her domestic abilities. Here I incorporate Fricker’s idea of hermeneutical injustice, which addresses when an agent lacks the social constructions and language to describe or identify their (harmful) experiences because of a ‘gap in collective interpretive resources’, and so cannot

[65] Jette, 1977, 23-24.
[66] Ibid., 17-18.
[67] Herzenberg and Howes, 1999, 15.
[68] Jette, 1977, 55-56. Herzenberg and Howes, 1999, 15.
[69] Brode, 1982, 64.
[70] Herzenberg and Howes, 1999, 126.

fully make sense of them.[71] Fricker identifies various ways hermeneutical injustices negatively affect an individual's psychology, going so far as to hinder self-development and prevent them 'from becoming who they are'.[72] Gender roles are significant in determining both the nature of a field and the types of jobs one pursues.[73] One could argue Jette may not have been fully able to self-actualise her desires, and was wronged in her capacity as a knower, because she did not possess the tools to understand how gender affected her experiences and desires. This also draws on the concept of adaptive preferences, wherein one's desires can be warped under harmful influences (e.g. it is easier to want to be a housewife in a society that values women as housewives over physicists).[74] It seems significant, that her reasons for not working are unrelated to confidence in scientific ability or concerns about belonging.

More interesting insights can be gleaned from Libby's narration of Laura Fermi. Libby had a self-confessed inability to understand Fermi's life path:

> '[f]or reasons that now seem inexplicable, Laura had been a student of general science at the University of Rome and secondarily had followed a course of electricity for engineers. But it seems that she shed any pretensions of interest in the sciences as soon as possible, namely, directly after passing the 2-year physics examination… It is puzzling why she had enrolled in science and engineering in Rome.'[75]

This seems to reflect the 'leaky pipeline' phenomenon that women are more likely to leave the field of physics at higher educational levels than men.[76] Libby referred to how Fermi did not seem to possess that natural curiosity previously discussed, writing '[i]n the first dozen years I knew her, she never asked a question about the physical world, never showed the least curiosity about the Manhattan District… She says she avoided any questions that might lead to scientific explanations'. Fermi herself barely discusses her degree in her book, *Atoms in the Family*.[77] One of Libby's comments is revealing as to how gendered influences may have affected Fermi's attitudes towards physics after her undergraduate degree: 'Laura says she had placed her husband on a pedestal and physics seems to have been on a pedestal also'.[78] As for many

[71] Fricker, 2007, 1.
[72] Ibid., 5.
[73] Blue et al, 2016, 5.
[74] E.g. Baber, 2007. Khader, 2011.
[75] Libby, 1979, 24-25.
[76] Ong, 2005, 596. Resmini, 2016.
[77] Fermi, 1961.
[78] Libby, 1979, 24. See Fermi, 1961, 58-60.

young women in the current day, elevating physics to this sort of hierarchical position can discourage them from participating within it.

However, Libby describes Fermi as 'intelligent but not interested in science… [she had] no trouble writing excellent descriptions of… experiments and their conclusions. This may prove that she could have been a scientist had she so chosen, which she certainly did not.'[79] This is an important line, suggesting key distinctions between a woman who is not very good at science, a woman who 'fears' science, and a woman who is not very *interested* in science. Fermi seems to have been self-assured in her own competence: 'she took a back seat to no one, and argued with [her husband Enrico] invincibly on any subject that she felt belonged to her domain'.[80] While it is good to engage more women in the physical sciences, and create environments for them to do so, we must also allow women the agency to not be interested .

Stereotypes and Identity

In this section, I aim to investigate women's relationships to their identities as women and as scientists, using the Project somewhat as a limiting case given the specific nature of its environments, and incorporating modern scholarship to demonstrate relevance for women in the US today.

I employ Fricker's definition of a stereotype as an association or 'cognitive commitment' between a social group and an attribute that is widely held, neutral as to its reliability, and may be positive or negative.[81] I also assume gender can be performative; gender identity arises partially via the repetition of certain acts, self-presentations, and stylisations, meaning gender is non-constant throughout time.[82] Fricker outlines 'stereotype threat', whereupon a member of a group subjected to a negative stereotype will tend towards self-fulfilling behaviour consistent with the stereotype.[83] Following this, it seems plausible the more a person identifies (in this sense of the word) as a woman, the more they will behave according to negative stereotypes surrounding woman as a category as they 'perform gender'– pertinently, that

[79] Libby, 1979, 30.
[80] Ibid., 26.
[81] Fricker, 2007, 30-31.
[82] Blue et al, 2016, 6.
[83] Fricker, 2007, 55-57.

women are not good at science. Research reflects this; test performances are adversely affected when the identity domains with negative stereotypes are emphasised for the individual.[84]

Gender and Appearance

Stereotypes are often expressed via visual imagery representing associations between a social group and an attribute. A paper by Blue et al. summarises our most common visual stereotype of a scientist as 'a white male with crazy hair, wearing a lab coat'.[85] Fricker has argued it is hardest to identify impact on judgement caused by visually embodied stereotypes, as we are often less consciously aware of how they condition epistemic decision making.[86] Research has shown the body can act as a vehicle for signalling scientific competence and membership via self-presentation, for example with women in physics who style themselves in more 'masculine' modes given strong masculine affiliations of the field as previously discussed.[87]

The Project was conducive to breakdowns of dress-related gender roles, most notably at Los Alamos. It tends to be housewives rather than female scientists who discussed this, although certainly the latter group made references to clothing and appearance.[88] Accounts written by men contain even fewer references. The most comprehensive summary of dress and its connotations I came across is the following passage, written by Charlie Masters, who taught at the Los Alamos school:

> 'Every soul… reflected the untensioned and uninibiting environment at least in superficial ways, if manner of dressing and grooming may be judged superficial… Professors who had always fretted under the collar-and-tie tradition of conservative classrooms now cast aside sartorial restraint and puttered happily about in ragged slacks, plaid shirts, and sheep-lined shortcoats. Women long hampered by the hat-and-glove requirements of campus towns now tramped to tea at friends' houses in flat-heeled moccasins, with bandana-wrapped heads, and threw overalled legs blissfully over the arms of GI chairs as they gossiped. Of course, it would have been too much to ask the ladies to forego the evening dresses they had cherished through all the vicissitudes of a

[84] Ito et al, 2016, 5.
[85] Blue et al, 2016, 7.
[86] Fricker, 2007, 36-37, 40.
[87] Ong, 2005, 594, 600-601, 605, 612. Staley, 2023, 14.
[88] E.g. Libby, 1979, 4.

> long overland trek or throughout a harassing overseas voyage… The evening dress was, perhaps, our dearest anomaly on the Hill. The men, however, luxuriating in clothes that were practical and comfortable and curiously, becoming, refused to match the splendor of their wives after dark'.[89]

Masters' writing conveys a clear sense of the freedom both men (denoted by the term 'Professor', which is a gender neutral term yet would have been an extremely unusual title for a woman) and women felt in discarding the restrictive clothes embodying restrictive societal roles. Jette related her initial horror at hearing women on the site wore blue jeans, and laughed at this after her eventual conversion.[90] Women – in contrast to men, according to Masters – still clearly enjoyed dressing up, but appreciated additional elements of practicality and associated freedoms. Masters was also explicit in relating this to the Los Alamos site.

Considering working wives as a social category, Serber outlined they only differed as they had 'less time for shopping and for housework', and in the 'trivial' ways of appearance: they looked tired (partly because of frequent outdoor activity), and 'dressed differently. We usually wore slacks, and sometimes jeans. We very often needed a shampoo. We weren't sloppy, really, but I am afraid we looked as if we dressed in a hurry, which we did'.[91] This is suggestive of the working women assuming homogenised and arguably masculinised modes of dress and presentation, which could be understood as increasing their sense of belonging. Interestingly, however, Serber followed this with '[a] new arrival on the Hill was jostled out of his conventional idea of what a library staff is supposed to look like. Here, he was apt to find the librarian in blue jeans, one assistant in the best sweater-girl tradition, another in strict WAC uniform, and a third in the latest Lane Bryant number… That was the thing about Los Alamos. It was anything but conventional'. This relates gender roles, the importance of outdoor activity, and unconventionality in terms of acceptance of difference at the Los Alamos site. Serber's attitude that her social group were not significantly different from other women suggests in this environment, she did not undergo fragmentation or actively suppress her feminine identity.

Writing in 1946, Wilson related dress to stereotypes and the media:

[89] Masters 1988, 119-121.
[90] Jette, 1977, 19, 90, 157.
[91] Serber, 1988, 70-71.

> ‘I find a misapprehension about scientists in current newspaper stories about them. “So and so,” says a newspaper reporter in surprise, “does not look like a typical scientist.” Had the reporter lived at Los Alamos for a while, he would have realized that there simply aren’t any typical scientists, that an almost flamboyant individualism is the rule…The people on the Hill were tolerant of idiosyncrasy. We were, if anything, proud of the fact that in science one need not conform to a pattern of dress of behaviour in order to succeed. I would have been happy to conduct a journalist down the dusty roads of Los Alamos while he sought that stereotype of his, the typical scientist… The only thing typical about the men of science at Los Alamos was that they were atypical’.[92]

Los Alamos, which at first glance would have been a much more difficult place for a woman to participate scientifically, was in this way unusually modern, as research today shows conforming to patterns of dress and behaviour are related to success rates in physics and perceptions of scientific competence.

Of course, there is some contradiction inherent in this analysis, given Wilson also only refers to ‘men of science’, meaning her recognition of the limits of stereotyping are still somewhat hindered by more rigid gender roles of the period. Of Oak Ridge, Weaver wrote ‘[i]ndeed, I’m ashamed to say that I occasionally heard men… discuss female scientists in insulting terms – concerning appearance, not competence – but that I did not rise to their defense’.[93] While this suggests women were not being judged on their scientific competence directly, perception of scientific competence can be deeply intertwined with physical appearance.

Social Order at Los Alamos

Many scientists working on the Project – particularly at Los Alamos – were in their mid-twenties, partly as nuclear physics was a relatively new field (especially in the US).[94] This, alongside their eccentricities, suggests why Los Alamos was not as entrenched in old social orders. The physical barrier of the fence around the site has been described by Marshak as having ‘a real effect on the psychology of the people behind it’ as a ‘tangible barrier’ keeping Los Alamos separate from the outside world.[95] Los Alamos had its own legal status, being part

[92] Wilson, 1988, 52-53.
[93] Weaver, 1999, viii.
[94] Allred, 1996, xiv. Marshak, 1988, 19.
[95] Marshak, 1988, 5.

of a military operation yet largely occupied with civilians.[96] The role of the army was significant in creating separation between Los Alamos and the 'Outer World' as distinct social groups.[97]

However, hierarchies certainly did exist on the site. One recurring suggestion in the sources is that this was determined by, in Marshak's words, 'the positions one's husband held in the Laboratory.'[98] Obviously, this is a highly gendered statement, and demonstrates both the status inherent in scientific roles and wife as a social category. Brode described how '[t]here was also a certain prestige involved in possessing a badge admitting one to the inner sanctum', i.e. the Technical Area, which had its own fence.[99] Jette was more dismissive, writing '[s]ome people deluded themselves with the idea the color of a badge indicated the individual's social standing', although in a sentiment echoed by Bacher does relate how the square dancing group at Los Alamos was a specific environment where badge-related hierarchies broke down.[100] Other evidence exists of class discrimination between scientists and physical labourers such as machinists and carpenters, and has been noted also of Oak Ridge.[101]

Los Alamos still presented as a masculine environment. For example, the only place in the hospital available for women was the maternity ward.[102] The guard hut for entry into the site was decorated with 'starkly naked pinup girls', not usually conducive to increasing a sense of belonging in women.[103] Brode related how Joseph Hirschfelder, the physicist who trained her as a computer, put Esquire calendars on the walls of his office for 'decoration and cheerfulness', in his words.[104] Ong has discussed how pin-up images in the lab can cause women to feel excluded and as if they don't belong, and 'reclaim the physics environment as a white, male (and heterosexual) space'.[105]

[96] Kimball Smith, 1988, 74-75.
[97] Wolkowitz, 2000, 246.
[98] Marshak, 1988, 9.
[99] Brode, 1982, 65. Roensch, 1993, 9.
[100] Bacher, 1988, 112. Jette, 1977, 156.
[101] E.g. Mark, 1988, 38. Wolkowitz, 2000, 241-242.
[102] Jette, 1977, 94.
[103] Marshak, 1988, 5.
[104] Brode, 1982, 64.
[105] Ong, 2005, 611.

Martyrs and Heroes

Historically, a greater proportion of scientists have undoubtably been men. Positive traits identified with good science are masculine-coded whereas feminine-coded positive traits are more typically associated with nurturing and caretaking.[106] One study found female physics graduate students frequently identified feminine traits negatively, and in contradiction to physics and logic.[107] This phenomenon is intertwined with historical and social conceptions of women as intuitive and emotionally intelligent, but irrational and illogical, making them unsuited to work in the physical sciences.[108]

Interestingly, in her aforementioned chapter 'Great Scientists', Rona somewhat unites the stereotypically feminine and masculine forms of intelligence (i.e. emotional versus logical): great scientists are 'able to… perceive in a visionary way, as in a dream… which can be developed by experiments and logical thinking'.[109] Her exemplars, however, are Marie Curie and Ernest Rutherford. Curie was commended for her 'long, assiduous, unbroken, labor', while Rutherford for his 'imagination', 'courage to carry out experiments without fear', and 'extraordinary confidence in his methods', as well as 'intense curiosity'. The gendered connotations of celebrating these different qualities in these different scientists are notable.

Research suggests the media is significant in spreading and perpetuating scientific stereotypes, in ways not mitigated by 'short-term interventions'.[110] Wilson's quotation about a reporter demonstrates existence and awareness of scientist stereotypes in the media around the time of the Project. Rona's writing suggests her image of Curie developed from the media. At the prospect of meeting Curie, Rona described:

> '[I had] misgivings… I believed that such a famous person… would live in an ivory tower and could be approached only by appointment. When I entered the laboratory, however, I saw a slight, graying lady in a lab coat working among the students… We were so impressed by Mme. Curie's ascetism… I admired Mme. Curie's modesty, and I thought

[106] Blue et al., 2016, 5.
[107] Ibid., 7.
[108] Ibid., 5. Fricker, 2007, 15, 31-32. Gender relationships with medicine and the biological sciences are somewhat more complicated and historically contingent, partly because of caretaking components to medical roles.
[109] Rona, 1978, 49.
[110] Blue et al, 2016, 7.

> of Einstein's remark: "Mme. Curie is the only person I know who is completely unspoiled by fame."… Rutherford attended one of the Solvay Conferences and wrote that Mme. Curie was also present in her modest and noble self'.[111]

Rona's description of Curie suggests her image of Curie as the scientific martyr existed before she met Curie, and that by the time she wrote her book, this image was perpetuated by writings of others.

This image of the 'modest' female scientist utterly committed to her scientific ideals stands somewhat in contrast with the white male genius stereotype (which Rona arguably plays into with her description of Rutherford).[112] Herzenberg and Howes described how '[t]he popular heroes of the project were a group of theoretical physicists, all male, whose arcane knowledge of quantum mechanics and the structure of the atomic nucleus first hinted at the possibility of a weapon based on nuclear fission'.[113] Relationships to these 'heroes' at Los Alamos were summarised neatly by Ruth Marshak, a teacher at the Los Alamos school married to physicist Robert Marshak: 'the mesa was Celebrity Land… Los Alamos stood for the same sort of thing that Hollywood represents to an aspiring starlet. Most of the great men of physics and chemistry were there at one time or another'.[114] There is no question of 'great men and women', despite Meitner and Curie looming large within the world of nuclear physics (although Curie had died by this time and Meitner refused to work on the atomic bomb).[115]

The most popular emergent figures of the Project are probably Robert Oppenheimer and Richard Feynman. Oppenheimer was director of the Los Alamos site during the War, while Feynman was a physicist and head of the computation group.[116] Oppenheimer's public image is well summarised by Libby via his apparently self-aware 'display of the fragile, ascetic, sensitive, tense intellectual'. Rona described Curie's ascetism, yet the connotations appear different for Curie the humble martyr versus Oppenheimer the dramatized mystic. This was a role Libby argued Oppenheimer could occupy 'because he had a big daddy there to play the daddy role, namely, Groves… Without Groves, Oppenheimer might have played the daddy part

[111] Rona, 1978, 23-24, 26.
[112] Ong, 2005, 597.
[113] Herzenberg and Howes, 1999, 6.
[114] Marshak, 1988, 11.
[115] Herzenberg and Howes, 1999, 198.
[116] My account has been informed by Thorpe, 2006.

himself, being calm, totally impassive, mystic and remote, but in a controlled way'.[117] Libby's description of the 'daddy' role emphasises how Oppenheimer occupied a position as a specifically masculine hero. According to Herzenberg and Howes, '[c]ertainly the military would have found it inconceivable to deal with a female Oppenheimer'.[118] This is a demonstration of gender relationships and power within the Project as a weapon-building endeavour as well as a scientific one. While women were able to occupy many scientific and technical roles, given the need for perceived authority, the heroic 'daddy' could only have been a man even across different forms of masculinity.

Public perceptions of scientists of the Manhattan Project were in flux after the War due to increasing awareness of the bomb.[119] Public response was sometimes damning, and journalist Robert Jungk described scientists 'as mythical figures, over-lifesize Titans… 'the gods from hell'".[120] The source material is much more humanising, and shows awareness of media depictions, for example in '[t]he so-called mad scientist turned out to be a well-balanced person, even when subjected to the tremendous problems and responsibilities involved in the making of the atomic bomb'.[121] Wilson also used the word 'ascetic' to describe Oppenheimer, although in a somewhat tongue-in-cheek way while recounting his bout of chicken pox.[122] This is a good example of how Oppenheimer – or 'Oppie' – is often discussed in the source material, showing awareness of his public image yet providing a more human (and humorous) portrayal.

Feynman can perhaps better be summarised as the stereotypical 'eccentric white male genius' within physics. There are multiple mentions of Feynman in the source material, showing he was a well-known figure at the Los Alamos site, but are mostly on how he and his wife tried to 'beat' the censors who checked the mail (e.g. 'a young theoretical physicist whose wife was desperately ill in a sanitorium sought to divert her by writing her letters in code').[123] In an interview with Herzenberg and Howes, Naomi Livesay French, a mathematician within the Theoretical Division at Los Alamos, said '[t]he machine operators didn't understand Feynman,

[117] Libby, 1979, 105.
[118] Herzenberg and Howes, 1999, 18.
[119] For example, the 1946 release of John Hersey's book *Hiroshima* in The New Yorker affected public opinion on the bomb via exposition of its horrifying destructive power.
[120] Allred, 1996, xiv. Marshak, 1988, 19. Wolkowitz, 2000, 237.
[121] Barnett, 1988, 101.
[122] Wilson, 1988, 50.
[123] Ibid., 45.

but they liked him and worked well with him'.[124] This accords strongly with the idea of Feynman as a mystical, elevated figure of physics, unknowable but admirable to the women who worked under him.

Stereotypes have been found to significantly affect engagement within different academic fields in various ways, including stereotype threat. A study by Cheryan et al. within computer science concluded that in relation to role models, women's sense of belonging was most adversely affected when the role model accorded with stereotypes of the typical computer scientist, rather than if the role model was male or female.[125] One way women have been discouraged from physics, then, is by overemphasis within historical accounts on the same few male 'heroes' and female 'martyrs', who have both formed and conformed to stereotypes of what a scientist is and should look like. The stereotype of the female scientific martyr sometimes dissuades women from engaging within physics, as it suggests they must correlate with this highly demanding stereotype to succeed. This is why historical accounts of multiple female scientists are important.[126] Herzenberg and Howes wrote stories of the Project influenced their decisions to become physicists, but they unquestioningly believed the bomb was created only by men.[127]

Husbands and Wives

Almost all of the women with technical and scientific roles stayed single or married scientists. Herzenberg and Howes discuss singledom as a consequence of the scientific devotion women felt they needed to succeed: 'the public and many male physicists refused to believe that women could handle physics. This view was surely a big factor behind the feeling of many women active in research that they had to remain single in order to dedicate themselves of science... Married female physicists were generally treated as talented amateurs, and were given lab space only if their husbands demanded it'.[128] They note 615 out of 687 women in the 1923 edition of *American Men of Science* – with acknowledgment of 'men' as the category for 'person' – were unmarried. Given the false yet socially somewhat real dichotomy between being able to marry and being a scientist, it is understandable many women chose the former.

[124] Herzenberg and Howes, 1999, 101-102, 106-107.
[125] Ito et al, 2007, 4.
[126] Fara, 2013.
[127] Herzenberg and Howes, 1999, 1-2.
[128] Ibid., 36.

Multiple sources refer to high rates of marriage between staff members.[129] Wolkowitz has identified 'husbands' and 'wives' as important social categories within women's accounts of the Project. Many of the women centralise their roles as wives within their accounts (and therefore, one can presume, their identities), although female physicists mentioned their roles as wives by far the least.[130] If within marriage, gender roles dictate the man must possess more social goods than the woman – status, power, money, education, strength, and so forth – it follows a woman with a technical or scientific education must marry a man with the same or higher level of education. Another benefit of a scientist husband, beyond common interests, is a strengthened social support network when able to discuss scientific problems and frustrations. The importance of social support outside of the classroom to allow women to persist in STEM has been identified within modern PER, as a coping resource from stress as well as a way to increase sense of belonging.[131] This phenomenon is somewhat complicated, therefore. While women having to 'marry up' is a phenomenon caused by rigid, unfair gender boundaries, which may discourage them from engaging with physics if concerned about being unable to find a partner, having a husband in science can positively increase a sense of belonging and social support.

Womanhood Versus Scientific Identity

In this subsection, I aim to explore relationships between feminine and scientific identity. Research has shown women in STEM fields today are frequently concerned about gender passing, and tension between gender performance and identity performance associated with scientific stereotypes causes women to leave the field or undergo fragmentation practices, the first being more likely.[132]

I again borrow from Ito et al. as they distinguish belonging from identification. The latter is defined as 'the degree to which a particular domain is an important element of one's self-definition'.[133] Identity is complex, and gender identity not the sole arbiter of identity and

[129] E.g. ibid., 137.
[130] Wolkowitz, 2000, 242.
[131] Ito et al, 2016, 6-7.
[132] Ong, 2005, 602, 604.
[133] Ito et al, 2016, 1-2.

personhood.[134] Within the material, it appears women working as scientists within the Project often felt more identification with 'scientist' than 'woman', much as female physicists identified less as 'wives' than other women at the site. The clearest example comes from a quotation given by Maria Goeppert Mayer to historian Joan Dash. Within the Project, Mayer worked mainly on isotope separation at Columbia University, also making several visits to the Los Alamos site.[135] She later won a Nobel Prize for work in nuclear physics, and Dash wrote the following of the Prize Ceremony:

> 'She thought especially of the physicist Max Born, who had been her teacher; she thought of her father… "He might have felt that his advice to me, 'Never become a woman!' had really come true." Her father had encouraged her to have a profession; don't just wait for some man, he used to tell her, for he had seen too many women dwindle into marriage, motherhood, baby tending and dullness.'[136]

Reference is made both to an inspiring teacher and encouraging father figure. Most significant about this quotation, however, is how it illustrates the conflict Mayer experienced between feminine and scientific identity, as well as the choice made (it is presented somewhat as a choice) to prioritise the latter. As Ong depicts for the present day, 'the achievement of "normal" or "ordinary" womanhood clearly stands at odds with the achievement of "normal" or "ordinary" success in scientific fields.'[137]

Pertinently, on a graduate supervisor, George Bredig, Rona wrote:

> '[f]rom time to time we were invited to his home, where his lovely wife served us tasty German cakes. The only drawback was that, because I was the only woman at the time in the laboratory, I had to join the ladies. I felt much out of place in this group. The conversations dealt with children, cooking preserves; recipes were exchanged. To these discussions I could not contribute. How I longed to be with my colleagues, to hear and talk shop'.[138]

In this case, Rona's society determined she belonged in the group denoted by identity component of 'woman', while Rona wishes to be categorised as 'scientist', and related she was

[134] Ibid., 8.
[135] Herzenberg and Howes, 1999, 37, 47.
[136] Dash, 1973, 233.
[137] Ong, 2005, 594.
[138] Rona, 1978, 4. Also in Herzenberg and Howes, 1999, 88.

unable to participate in normal modes of womanhood for the period. There is a harmful disconnect between Rona's self-identifications and identifications imposed upon her.

If one does not 'become a woman', does one 'become a man'? Given gender is in some ways a performance, then presumably, if one decreases the frequency of 'female' acts and increases the frequency of 'male' acts – particularly in a society within which female and male are strictly defined – then one is undergoing some form of gender transformation. For example, a study on gender and engineering found certain women attempted (to varying degrees of consciousness) to 'act like men', for example by pursuing reputations of high competence, acting like 'one of the boys', and becoming 'antiwoman'.[139]

However, to suggest women who 'act like men' – fragmentation or otherwise – engage better does not present the full picture. Female scientists at the Manhattan Project, like Rona at the tea party, were judged both as scientists and as women. Studies from the 1980s onwards have demonstrated how self-promotion, assurance in one's own competence, or 'dominating' behaviour (often viewed as positive qualities in scientists) exhibited in women are viewed negatively, especially within scientific contexts.[140] Navigating gender roles within science is complex, as there is an expectation for women to behave 'like women' but for scientists to behave 'not like women'. Let me turn, for example, to a quotation from Rona on Irène Curie and her character: 'her disinclination toward gossip and small talk, her extreme seriousness, her reserve, and her disregard of her appearance'. [141] These traits are arguably masculine-coded. Notably, Rona wrote Curie's 'lady colleagues' disapproved of these things, recognising a gendered dimension to this disapproval from women who, perhaps, occupied more traditional gender roles.

It is interesting to consider Rona's discussions of Meitner, as an example of a successful female scientist without the level of self-assurance I have discussed of women like Libby: 'even after a long scientific career, she had continued to experience stage fright… Her voice quivered a bit when she started to speak, but soon an eloquency became apparent. Once she became caught up in her scientific theme, she could deliver an interesting and clear lecture'.[142] Perhaps we can

[139] Blue et al, 2016, 7.
[140] Ong, 2005, 598-599.
[141] Rona, 1978, 27.
[142] Ibid., 13.

understand this as Meitner succeeding at the top levels of physics while retaining a more typically feminine (i.e. shy) character due to conviction and love for science and intellectual courage, acting as more significant factors than ability to present with stereotypically masculine traits. Women at the Manhattan Project participated in various roles regardless of gender identification and expression, although certain modes increased sense of belonging within science and eased participation.

Encountering and Deciphering Sexism at the Manhattan Project

In this section, I examine narration of gendered experiences. There are various examples of outright gender discrimination. The prevailing attitude of the era was that women would 'obviously' be paid less than men.[143] Serber lamented the arbitrary nature of salaries for the 'working wife', as they had 'no bargaining power', a reflection of US society more broadly. From Fricker, I roughly define testimonial injustice as injustice that occurs when an individual is harmed in their capacity as a knower because of a deflated level of credibility being given to them on the basis of some identity prejudice. Various testimonial injustices related to the Project include women being excluded from publications, and facing disbelieving, 'bitter' reactions they could perform certain scientific tasks.[144] Most examples of sexism that are treated as such, however, are within the secondary literature. Discussions of clearly gendered experiences are rarely couched in language of feminist theory or misogyny, even in accounts written post the advent of second wave feminism. I found no outright condemnation of sexist behaviour.

While at Hanford, Libby, recognising how precipitate collects in the centre of swirling fluid and possible negative effects in the context of plutonium separation, was 'unable to convince [her bosses] to take this problem seriously, and indeed it did not give Hanford trouble. However, years later, after the war, there was a radiation accident caused by this effect.'[145] Was this sexism? Epistemic injustice? Libby doesn't comment. Similarly, on creating the pile at the University of Chicago, 'Walter Zinn forbade me to help with construction because he said he didn't want to discover he had been cussing out a girl when I took off my dust mask and cap…

[143] Herzenberg and Howes, 1999, 81.
[144] E.g. Ibid., 58-59, 75, 169.
[145] Libby, 1979, 172.

It was just as well because there was plenty for me to do'.[146] Libby appears pragmatic and unbothered about this interaction. However, she was excluded from normal activity for a scientist in her group due to her gender. This is also an interesting example of neutral dress being insufficiently obscuring to overcome gender roles, and detrimental to participation due to an individual prioritising Libby's gender identity over her scientific identity. Typically, one would wish to attribute Libby the agency to describe her own experiences as sexist and harmful or otherwise. Libby's book was published in 1979, meaning more tools of feminist theory within the social collective consciousness did exist, yet this does not necessarily mean she was highly aware of or able to access them. Analysing these sorts of interactions in terms of gender discrimination can be difficult even today. In this sense, gendered experiences like this may have been not just sexist, but also instances of hermeneutical injustice.

Rona made no comment on her father's opinion that medicine was too hard for a woman.[147] She 'respected his intentions' as she pursued physical chemistry. Similarly, Rona does not offer direct condemnation of Bredig for her repeated exclusion from the group of male physicists, although mentioned 'Fajans gave many small parties in the laboratory with no discrimination against women'.[148] Perhaps the contrast here is telling. Perhaps Libby charitably interpreted his wish not to swear in front of a woman as well-intentioned, particularly considering the role of hermeneutical injustice. One can recognise condemning agents who committed sexist acts is not always entirely reasonable, as these agents also did not possess as many social tools.[149] Historical context arguably mitigates the agent's responsibility to overcome deeply ingrained epistemic behaviours, akin to moral luck and condemnation of ethical judgements.[150]

While many historical accounts have side-lined women, women often diminished their own contributions. Even the title *Standing By and Making Do* perpetuates this idea of passivity. In most of the autobiographical material, the category 'scientist' at the Manhattan Project is gendered with male terms: '[t]o our husbands and to all the men who made the atomic bomb a reality this book is affectionately dedicated', '[w]e never spoke of anyone by the term Doctor or Professor, but always called him Mister', and so forth.[151] Women can also perpetrate

[146] Ibid., 86. See also Fermi, 1961, 193.
[147] Rona, 1978, 2.
[148] Ibid., 4.
[149] Fricker, 2007, 7, 14-15.
[150] Ibid., 33, 42.
[151] McKibbin, 1988, 25. Serber and Wilson, 1988, prefatory matter.

epistemic injustices against other women. Libby recounted an argument with Laura Fermi on nuclear power, which led to the following exchange: '"Well," [Laura Fermi and Freda Urey, wife of physicist Harold Urey] said, "we don't know that but we'll ask Harold when he comes home." To them, I am still 23 years old'.[152] Wolkowitz analyses this exchange in terms of a 'longstanding antagonism'. Fermi's own description of Libby is positive, saying she 'could do a man's job and do it well', but that 'because I refused… to smash atoms… she looked down on me'.[153] I argue this dismissal can still be understood as testimonial injustice.[154] Libby also suggested of Fermi 'she is a firm believer in a double standard of behavior for men and women'.[155] Can we interpret this to say Libby the physicist disagreed with Fermi the housewife in this regard? Libby does not offer further comment, but it seems plausible.

Returning to Rona's comment on 'lady colleagues', sexism and gender expectations caused tensions amongst women in ways not seem to have been experienced by men. Herzenberg and Howes have drawn attention to wariness between housewives and working women across sites, 'even at Los Alamos' where these two groups had greater overlap.[156] This wariness is shown by Jette's description of Hinton:

> 'Joan Hinton… played well and joined the music lovers. She was a congenital twitcher, human fly type. The night she first played at our apartment, she wore a skirt and did everything but climb the walls… I asked, "Does that girl ever sit still? Does she behave that way at work?" "When you're talking with her, she's just as likely to end up on the back of a chair with her feet on the seat as not," Eric admitted. "She's a smart cookie, but she had a progressive education." "Too progressive, if you ask me. I hope she wears slacks to work." "Not always; it gets pretty embarrassing."'[157]

Jette here expressed direct disapproval of Hinton's dress, and the word 'progressive' seems coded as extending too far outside of conventional gender roles, even for Los Alamos.

Epistemic injustice was also perpetuated against indigenous women at the Los Alamos site. Native American women were employed in fairly large numbers as cleaners and childminders

[152] Libby, 1979, 130.
[153] Fermi, 1961, 179.
[154] Wolkowitz, 2000, 243.
[155] Libby, 1979, 25.
[156] Herzenberg and Howes, 1999, 154-155.
[157] Jette, 1977, 118. Herzenberg and Howes, 1999, 154-155.

at Los Alamos, particularly as many wives living onsite were working.[158] Wolkowitz describes how in many accounts, 'they tend to be portrayed as exotic 'others' rather than as part of the core community'.[159] This is well conveyed by Masters describing a 'hilarious' pantomime skit put on by the British Mission at Los Alamos at their farewell party:

> '[t]he wife, harassed by many duties, employs a maid to do the housework. There enters, as the maid, a short, stolid figure shapelessly draped in a gaudy blanket… The figure moves with the pace of a bored snail… [and] turns a glum and owlish eye upon the audience. The crowd breaks into shouts of joy, recognizing Otto Frisch, one of the senior scientists of the British Mission. The maid listens expressionlessly to the detailed instructions of the housewife, who then departs in innocent trust. Left alone, the redskin domestic stalks idly but majestically about the room… she picks up a dirty dish from the heap, weighs it speculatively in her hand, and flings it to the floor... Next, she picks up a few more dishes and gingerly wipes their edges on the bottom of the window curtains. Her work is done. As recompense for her toil, she snitches a bottle from the refrigerator and swigs its contents appreciatively. When the mistress of the house returns, the maid impassively collects her fee and goes out, leaving the lady fuming with impotent rage'.[160]

Masters acknowledges this is exaggerated, but she also suggests there is truth in it. The contrast depicted between the 'innocent' white housewife and the 'ignorant', thieving, 'stolid' maid is stark. This account plays into stereotypes of the 'noble savage'.[161] This is a highly dehumanising portrayal, cruelly undermining their epistemic status and using what is now more widely recognised as a harmful slur.. While Los Alamos may have constituted a micro-society allowing certain specific breakdowns of normal gender roles, it still had 'in' and 'out' groups that led to epistemic injustices. It is similarly worth mentioning the distinct struggles faced by Black men and women at sites of the Manhattan Project. For example, African-American married couples were not allowed to live together at Oak Ridge.[162] Race, class, and their intersections with gender within the sites of Manhattan Project, and the related (particularly epistemic) injustices, demand further analysis.

[158] Serber, 1988, 68. Wolkowitz, 2000, 242.
[159] Wolkowitz, 2000, 242.
[160] Masters, 1988, 122-123.
[161] Ellingson, 2001.
[162] Wolkowitz, 2000, 241-242.

Epistemic Exclusion

First described by Frantz Fanon, psychic alienation is the harmful phenomenon that occurs when an individual is separated from some essential aspect of personhood. Fricker argues one essential attribute of personhood is ability to participate in testimonial knowledge spreading and 'enjoy the respect' that can be gleaned from epistemic trust.[163] Exclusion on the basis of social group is therefore epistemically as well as ethically harmful. Ong draws attention to how collaboration has been a historically feminine practice, yet within science, collaborative research methods often exclude women while helping men. For example, one study found women were often dominated by male peers, were more likely to take on administrative work, or experienced fears their fellow participants were more knowledgeable.[164] Rona's experiences at Bredig's parties exemplify harmful epistemic exclusion on the basis of social group, which subjected her to psychic alienation. I note she was excluded from a *scientific* epistemic network, as Rona discussed knowledge exchange amongst women – exchanging of recipes in particular – although also that she struggled to participate therein. I acknowledge this form of epistemic exchange as valuable, particularly as sharing knowledge within domestic spheres has constituted much of female epistemic exchange historically. Notably, however, Rona was harmed as she was prevented from participation in testimonial knowledge spreading.

Many accounts refer to female epistemic networks, such as '[i]n the evening, while the men who were at the test slept, and their children were tucked into bed, their women gathered at various places to discuss the events of the day'; however, there are almost no references to scientific knowledge exchange between women.[165] Weaver wrote:

> 'I don't recall any other female scientists working in my groups, and I didn't get to know any of the women working elsewhere at the site… My buddies were all men... I remember no sense of camaraderie among women; indeed, we viewed other women with some suspicion. The support that women give one another today did not exist then…When I read the personal reminiscences of the men who were nuclear pioneers, I'm struck by the importance they place on their friends and associates, and on the often intense interaction among them, which could lead to major insights in both the theory and practice of

[163] Fricker, 2007, 58-59.
[164] Ong, 2004, 599.
[165] Jette, 1977, 202.

science. And I'm a bit jealous. By and large, women did not share in that give-and-take'.[166]

Weaver draws attention to support amongst women at the time of writing in the late 1990s, which she felt did not exist in the 1940s before second-wave feminism, which appears somewhat an acknowledgement of hermeneutical injustice. This is a fascinating example, therefore, of gained social tools for women in science to describe and understand their experiences. Weaver's writing shows awareness of how women's epistemic exclusion and lack of equivalent epistemic networks hinders scientific success.

Many of the women working within the Project were unaware of what, exactly, they were working on.[167] Jette, on the first test of an atomic bomb, wrote '[the women] chortled with unholy glee when we heard that [General Groves] cautioned a security officer to keep the result of the test SECRET, particularly from the wives at Los Alamos… We… wondered how [Groves] could expect to keep anything seen over such a wide area secret, were particularly from the wives at Los Alamos.'[168] Here Jette asserts their skill at information gathering and exchange, yet coded as gossip, this can be understood as a self-asserted statement they are not epistemically trustworthy, which can be understood as psychic alienation. Gossip as epistemic exchange is strongly associated with femininity, and has a negative valence. I found no particular evidence of women defending themselves from this charge, but some degree of agreement– Wilson described herself as 'by nature blabbermouthed'.[169] Elsewhere, Jette seemed keen to assert her status as a knower, referring multiple times to having worked out the scientists in the Technical Area were working on a bomb despite no information from her husband.'[170] The sources suggest wives were excluded from knowing about their husbands' work because of tendency to gossip.[171] Marshak used emotive terms to describe this, having 'plied' her husband with many questions that 'quivered on [her] lips' which remained 'steadfastly' unanswered.[172] Clearly, this caused Marshak distress, and she described the

[166] Weaver, 1999, vii-viii. Herzenberg and Howes, 1999, 137.
[167] Herzenberg and Howes, 1999, 137-139, 148, 169. McKibbin, 1988, 27. Rona, 1978, 54-55.
[168] Jette, 1977, 198.
[169] Wilson, 1988, 45.
[170] Ibid., 108, 162, 182-183.
[171] Herzenberg and Howes, 1999, 16. Wilson, 1988, 43.
[172] Ibid., 51. Marshak, 1988, 1-2.

'loneliness and heartache' experienced by the wives of many scientists.[173] Gendered epistemic exclusion caused harm to women at the Project in multiple capacities.

Conclusion

Economic and military considerations may be less than ideal motivations to increase numbers of women in STEM. However, the Project did result in several women accumulating skills and starting scientific careers they most likely would have never been able to otherwise.[174] Ultimately, however, it did not have a large longstanding impact in terms of increasing numbers of women in the sciences, partly due to little attention paid to women at the Project in the media and literature.

Extending modern research to empirically analyse understudied source material, I have illustrated how despite major changes in physics culture and US society, surprisingly little has changed for women in terms of relationships between feminine and scientific identity. Firstly, this involved examining common factors for increasing women's engagement visible both in modern research and the sources. I argued understanding these is important to create environments and cultures inclusive of women, but overemphasis can be similarly problematic. In the second section, I explored relationships to social categories of 'woman', 'wife', and 'scientist'. Female physicists often experienced tensions between them and prioritised identification with the latter, while perceived primarily as the former. Examining related stereotypes, I discussed how feminine-coded 'martyr' and masculine-coded 'hero' stereotypes can be harmful. This also involved looking at gender within the specific environments of the Manhattan Project, for example as specificities of dress exemplify increased belonging and breakdown of typical gender roles. Within the final section, I considered varied instances of gender-related epistemic injustice within the field and the Project, and their associated harms and effects. I hope to have conveyed and partially unravelled the complexity of women's relationships with the physical sciences at the Manhattan Project and in the current day.

[173] Marshak, 1988, 11. In Wolkowitz, 2000, 243-244.
[174] Herzenberg and Howes, 1999, 17, 136, 141-142.

Bibliography

AAMC, Association of American Medical Colleges. (2022) *Applicants, First-Time Applicants, Acceptees, and Matriculants to U.S. MD-Granting Medical Schools by Gender, 2013-2014 through 2022-2023*. Online. Available at: https://www.aamc.org/media/9576/download?attachment [Accessed: 24/05/2023]

Allred, J. C. (1996) 'Preface' in *Behind Tall Fences: Stories and Experiences About Los Alamos at its Beginning*, pp. xiii – xiv. Los Alamos: Los Alamos Historical Society.

Baber, H. E. (2007) 'Adaptive Preference', *Social Theory and Practice*, **33**(1); pp. 105 – 126.

Bacher, J. (1988) 'Fresh Air and Alcohol' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 102 – 116. Los Alamos: Los Alamos Historical Society.

Barnett, S. B. (1988) 'Operation Los Alamos' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 88 – 101. Los Alamos: Los Alamos Historical Society..

Blue, J., Barthelemy, R., Cid, X. C. and Traxler, A. L. (2016) 'Enriching Gender in Physics Education Research: A Binary Past and a Complex Future', *Physical Review Physics Education Research*, **12**(2); 020114. Online. Available at: https://doi.org/10.1103/PhysRevPhysEducRes.12.020114 [Accessed: 02/05/2023]

Brode, B. (1982) 'Tales of Los Alamos', *Annals of the History of Computing*, **4**(1); pp. 63 – 65.

Dash, J. (1973) *A Life of One's Own: Three Gifted Women and the Men They Married*. New York: Harper & Row.

De Beauvoir, S. (1953) *The Second Sex*. London: Jonathan Cape Ltd.

Ellingson, T. (2001) *The Myth of the Noble Savage*. Los Angeles: University of California Press.

Fara, P. (2013) 'Women in Science: Weird Sisters?', *Nature*, **495**; pp. 43 – 44.

Fermi, L. (1961) *Atoms in the Family: My Life with Enrico Fermi*. Chicago: University of Chicago Press.

Feynman, R. (1982) 'Los Alamos from Below', *Annals of the History of Computing*, **4**(1); pp. 61 – 63.

Fricker, M. (2007) *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford: Oxford University Press.

Gailar, J. (1991) *Oak Ridge and Me: From Youth to Maturity*. Oak Ridge: Children's Museum of Oak Ridge.

Galison, P. (1997) *Image and Logic: A Material Culture of Microphysics*. Chicago: University of Chicago Press.

Groves, L. (1945) 'Foreword' in Smyth, H. D. *Atomic Energy for Military Purposes: The Official Report on the Development of the Atomic Bomb under the Auspices of the United States Government, 1940-1945*, p. v. New York: Carey Press Corporation.

Hersey, J. (1946) *Hiroshima*. New York: Alfred A. Knopf.

Herzenberg, C. and Howes, R. (1999) *Their Day in the Sun: Women of the Manhattan Project*. Philadelphia: Temple University Press.

Ito, T. A., Finkelstein, N. D., Lewis, K. L., Pollock, S. J., and Stout, J. G. (2016) 'Fitting In or Opting Out: A Review of Key Social-Psychological Factors Influencing a Sense of Belonging for Women in Physics', *Physical Review Physics Education Research*, **12**;

020110. Online. Available at: https://journals.aps.org/prper/abstract/10.1103/PhysRevPhysEducRes.12.020110 [Accessed: 04/05/2023]

Jette, E. (1977) *Inside Box 1663*. Los Alamos: Los Alamos Historical Society.

Khader, S. J. (2011) *Adaptive Preferences and Women's Empowerment*. Oxford: Oxford University Press.

Kimball Smith, A. (1988) 'Law and Order' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 72 – 87. Los Alamos: Los Alamos Historical Society.

Libby, Leona Marshall. (1979) *The Uranium People*. New York: Crane Russak.

Lyon, L. L. (1988) 'Foreword' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. v – vii. Los Alamos: Los Alamos Historical Society.

Mark, K. (1988) 'A Roof Over Our Heads' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 28 – 41. Los Alamos: Los Alamos Historical Society.

Marshak, R. (1988) 'Secret City' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 1 – 19. Los Alamos: Los Alamos Historical Society.

Masters, C. (1988) 'Going Native' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 116 – 130. Los Alamos: Los Alamos Historical Society.

McKibbin, D. (1988) '109 East Palace' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 20 – 27. Los Alamos: Los Alamos Historical Society.

Montmarquet, J. A. (1993) *Epistemic Virtue and Doxastic Responsibility*. Lanham, Maryland: Rowman and Littlefield.

Ong, M. (2005) 'Body Projects of Young Women of Color in Physics: Intersections of Gender, Race, and Science', *Social Problems*, **52**(4); pp. 593 – 617.

Pollard, W. G. (1978) 'Foreword' in Rona, E. *How It Came About: Radioactivity, Nuclear Physics, Atomic Energy*, pp. i – ii. Oak Ridge, Tennessee: Oak Ridge Associated Universities.

Resmini, M. (2016) 'The 'Leaky Pipeline'', *Chemistry: A European Journal*, **22**; pp. 3553 – 3554.

Roensch, E. (J). S. (1993) *Life Within Limits: Glimpses of Everyday Life at Los Alamos, New Mexico, Seen Through the Experiences of a Young Female Soldier While on Military Service There, May 1944 to April 1946*. Los Alamos: Los Alamos Historical Society.

Rona, E. (1978) *How It Came About: Radioactivity, Nuclear Physics, Atomic Energy*. Oak Ridge, Tennessee: Oak Ridge Associated Universities.

Sax, L. J. Barthelemy, R. S. Kathleen, J. Lim, G. (2016) 'Women in Physics: A Comparison to Science, Technology, Engineering, and Math Education over Four Decades', *Physics Review Education Review*, **12**(2); 020108. Online. Available at: https://journals.aps.org/prper/abstract/10.1103/PhysRevPhysEducRes.12.020108 [Accessed: 18/05/2023]

Schwarz, R. (2008) *The Making of the History of the Atomic Bomb: Henry DeWolf Smyth and the Historiography of the Manhattan Project*. [Doctoral dissertation, Princeton University] Princeton: Princeton University ProQuest Dissertations Publishing.

Serber, C. (1988) 'Labor Pains' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 56 – 71. Los Alamos: Los Alamos Historical Society.

Serber, C. and Wilson, J. S. (1988) *Standing By and Making Do: Women of Wartime Los Alamos*. Los Alamos: Los Alamos Historical Society.

Smyth, H. D. (1945) *Atomic Energy for Military Purposes: The Official Report on the Development of the Atomic Bomb under the Auspices of the United States Government, 1940-1945*. New York: Carey Press Corporation.

Staley, R. (2023) 'Physics as a Human Endeavor' in Heron, P. and Taşar, M. F. *The International Handbook of Physic Education Research: Physics Education Research Special Topics*, pp. 6-1–6-26. Melville, New York: AIP Publishing,

Thorpe, C. (2006) *Oppenheimer: The Tragic Intellect*. Chicago: University of Chicago Press.

Weaver, E. C. (1999) 'Foreword' in Herzenberg, C. and Howes, R. *Their Day in the Sun: Women of the Manhattan Project*, pp. vii – viii. Philadelphia: Temple University Press.

Wellerstein, A. (2008) 'Patenting the Bomb: Nuclear Weapons, Intellectual Property, and Technological Control', *Isis*, **99**; pp. 57 – 87.

Wilson, J. S. (1988) 'Introduction' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. x – xi. Los Alamos: Los Alamos Historical Society.

Wilson, J. S. (1988) 'Not Quite Eden' in Serber, C. and Wilson, J. S. (eds.) *Standing By and Making Do: Women of Wartime Los Alamos*, pp. 42 – 55. Los Alamos: Los Alamos Historical Society.

Wolkowitz, C. (2000) 'Nuclear Families: Women's Narratives of the Making of the Atomic Bomb' in Ahmed, S., Kilby, J., Lury, C., McNeil, M. and Skeggs, B. (eds.) *Transformations: Thinking Through Feminism*, pp. 235 – 249. Boca Raton, Florida: Routledge.